Knowledge Is Power (KIP): Software Requirements Specification

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1. Introduction

1.1. Purpose

The following document defines the details and specifications of Knowledge Is Power (KIP), a legal learning platform that uses interactive games and a progress-centered learning cycle to improve young adults' legal literacy. It establishes the functional and nonfunctional requirements that direct the system's development, testing, evaluation, and maintenance.

1.2. Scope

KIP is a web-based legal learning platform designed to provide interactive educational content, track users' learning progress, offer source-backed feedback, and adapt presented material based on their activity and demonstrated understanding. The platform is intended to be publicly accessible on desktop and mobile devices and to support continued learning across multiple sessions.

KIP offers legal content focused exclusively on U.S. federal and state law. It will not attempt to exhaustively cover all areas of U.S. law, instead prioritizing topics most relevant to young adults, who represent the platform's primary target audience. Furthermore, the legal content provided by KIP is not intended to constitute professional legal advice or substitute for formal legal education. The platform will not grant certifications, licenses, degrees, or other formal legal or educational credentials.

KIP operates within the application area of legal literacy and game-based education. Young adults regularly encounter laws affecting areas such as employment, housing, consumer activity, interactions with law enforcement, and personal rights, yet the information needed to understand

these laws is often distributed across complex legal sources and presented in forms intended for trained legal professionals rather than general audiences. This creates a barrier between the existence of legal information and an individual's ability to understand, retain, and apply it in relevant situations.

KIP aims to reduce this barrier by presenting selected legal concepts through accessible, interactive learning experiences. Its objectives are to improve users' awareness and retention of commonly applicable laws, identify areas of weak understanding through continued assessment, provide feedback grounded in authoritative legal sources, and encourage users to engage more directly with those sources when further information is needed.

1.3. Definitions, Acronyms, and Abbreviations

KIP: Knowledge Is Power

1.4. References

N/A

1.5. Overview

Section 1 first establishes the context in which KIP operates and the roles its major components play within the platform. The remainder of this document describes KIP from progressively more detailed perspectives.

Section 2 then provides an overall description of the product, including its relationship to existing educational platforms, major functions, and intended users.

Section 3 finally defines the system requirements that govern KIP's implementation. It divides the platform into its major system features and specifies the purpose, expected stimulus and response behavior, and functional requirements associated with each feature. These requirements cover the user-facing interface, account and administrative functionality, game operation, feedback and progress tracking, content curation, legal source referencing, and supporting data systems.

2. Overall Description

2.1. Product Perspective

KIP shares some characteristics with existing online education platforms such as Quizlet, including interactive activities and performance tracking. However, while such platforms often emphasize rote memorization and recall, KIP uses scenario-based games to evaluate a user's ability to apply legal concepts in realistic and relatable situations.

This approach is particularly well suited to legal education because many legal concepts can be understood in terms of actions, conditions, and consequences. KIP can therefore present users with simulated real-world legal scenarios, accept their decisions as input, and demonstrate the resulting legal outcomes, so that the platform can assess not only whether a user has remembered legal information, but also whether they have recognized and applied it in context.

KIP is not the first product to apply interactive learning to legal education. Existing platforms such as Street Law, iCivics, and Law Dojo provide legal or civic learning experiences that overlap with KIP's intended functionality. KIP differentiates itself through responsive user

feedback, progress-sensitive content recommendations, and support for state-specific legal content.

2.2. Product Functions

The capabilities of KIP described above are supported by several major functional components of the system. Together, these components coordinate to form a continuous learning cycle in which users interact with educational content, receive feedback on their decisions, accumulate a record of their understanding, and are subsequently presented with material appropriate to their demonstrated progress.

Each component is responsible for a distinct part of this process while exchanging information with other components where necessary. At a high level, KIP consists of the following major functional components:

- Authentication System — identifies users and maintains access to their persistent data.
- Feedback System — evaluates user choices to relay relevant explanation and feedback.
- Progress Tracker — records user activity, topic proficiency, and concept comprehension.
- Curation System — prepares content according to user progress and understanding.
- Game Engine — manages active legal games, user choices, and resulting outcomes.
- Source Reference System — associates content and feedback with official legal sources.

2.3. User Characteristics

KIP's user base can be divided into two broad categories: internal and external users. Internal users are individuals with administrative access to KIP's live system and content-management

functions. These users are responsible for creating and maintaining educational content, introducing new legal topics, and reviewing legal source references to ensure that the information presented by the platform remains accurate and current.

External users are individuals who access KIP primarily as learners or consumers of its educational content. This group includes young adults using the platform independently, students using it to enhance their studies, and educators using it as a supplementary instructional resource. External users are expected to vary in their prior legal knowledge, educational background, and motivations for using the platform.

2.4. Constraints

N/A

2.5. Assumptions and Dependencies

N/A

3. System Requirements

KIP shall contain 15 main system features:

- Navbar
- Dashboard
- Browser
- Account Portal
- User Settings
- Admin Panel
- Sitewide Theme
- Game Window
- Game Engine
- Progress Tracking
- Curation
- Feedback
- Legal Reference Algorithm
- Data Initialization Layer and Metric Tracking
- User Profile and Match Metrics Initialization

3.1. System Features

All system features coordinate together to facilitate KIP’s essential functionality.

3.1.1. Navbar

3.1.1.1. Introduction/Purpose of Feature

The Navbar is displayed on each page of KIP and provides persistent access to the application's primary pages.

3.1.1.2. Stimulus/Response (O: Macaluso)

Stimulus: The user activates a Navbar entry.

Response: The system displays the page associated with the entry. The entry for the displayed page is visually different from the other entries.

3.1.1.3. Functional Requirements

3.1.1.3.1. Navbar Entries (O: Elmousalami)

The system shall display the following Navbar entries to an authenticated user: Dashboard, Games, Progress, Settings.

3.1.1.3.2. Navbar Destination (O: Elmousalami)

The system shall visually display the page associated with a Navbar entry when the user clicks on that entry.

3.1.1.3.3. Site Branding (O: Noor)

The system shall display the KIP name and logo in the Navbar throughout the application.

3.1.2. Dashboard

3.1.2.1. Introduction/Purpose of Feature

The Dashboard is the first page displayed to an authenticated user of KIP upon logging in, offering them a welcome message and recommended game. For users with past activity, it additionally displays a profile summary and recent game sessions.

3.1.2.2. Stimulus/Response

Stimulus: The user displays the Dashboard page.

Response: The system displays the welcome message and recommended game. The system also displays Recent Activity and Profile Summary to an authenticated user.

3.1.2.3. Functional Requirements

3.1.2.3.1. Daily Recommendation Display

The system shall display a recommended game on the Dashboard page.

3.1.2.3.2. Daily Recommendation Start Playing (O: Elmousalami)

The system shall display the browser when the user clicks the Start Playing button.

3.1.2.3.3. Recent Activity Visibility (O: Elmousalami)

The system shall display Recent Activity only to an authenticated user.

3.1.2.3.4. Recent Activity Entry (O: Elmousalami)

The system shall display the following for each recent game session: Game name, Topic, Time elapsed since the game session, Score as a percentage.

3.1.2.3.5. Recent Activity Score Color (O: Elmousalami, M1: Keckley)

The system shall display the performance score of each recorded game session with a color derived from the percentage of prompts the user responded correctly to:

- 0-25%: Red
- 26-50%: Orange
- 51-75%: Yellow
- 76-100%: Green

3.1.2.3.6. Recent Activity Empty State (O: Elmousalami)

The system shall display a message to an authenticated user who has no recent game sessions.

3.1.3. Browser

3.1.3.1. Introduction/Purpose of Feature

The Browser lists the games available in KIP. It gives users the information needed to choose a game.

3.1.3.2. Stimulus/Response

Stimulus: The user displays the browser.

Response: The system displays each available game with its details.

3.1.3.3. Functional Requirements

3.1.3.3.1. Game List (O: Elmousalami)

The system shall display every available game on the Browser.

3.1.3.3.2. Game Details (O: Elmousalami)

The system shall display the following for each game: Name, Description, Status, Experience.

3.1.3.3.3. Topic List (O: Noor)

The system shall display all available legal topics within the Browser.

3.1.3.3.4. Topic Selection (O: Noor)

The system shall allow the user to select a legal topic from the topic list.

3.1.3.3.5. Topic Filtering (O: Noor)

The system shall display only the games associated with the legal topic selected by the user.

3.1.3.3.6. Game Card (O: Noor)

The system shall display each available game using a reusable game card within the Browser.

3.1.3.3.7. Game Statistics (O: Noor)

The system shall display each game's status and experience statistics based on the user's progress.

3.1.3.3.8. Game Card Information (O: Noor)

The system shall display the game title, thumbnail, description, history statistic, and experience statistic on each game card.

3.1.4. Account Portal

3.1.4.1. Introduction/Purpose of Feature

The Account Portal allows a registered user to access KIP within the context of their account's activity and preferences.

3.1.4.2. Stimulus/Response

Stimulus: The unauthenticated user visits the KIP website and clicks on the Account Portal button.

Response: The system displays the Account Portal page.

3.1.4.3. Associated Functional Requirements

3.1.4.3.1. Log In (O: Hoar, M1: Keckley)

The system shall allow a registered user to log in using their account credentials.

3.1.4.3.2. Sign Out (O: Hoar, M1: Keckley)

The system shall allow an authenticated user to sign out.

3.1.4.3.3. Access Control (O: Hoar, M1: Keckley)

The system shall restrict administrative functionality to authenticated admin accounts.

3.1.5. User Settings

3.1.5.1. Introduction/Purpose of Feature

User Settings allows a learner to configure various settings.

3.1.5.2. Stimulus/Response

Stimulus: The learner clicks the User Settings button.

Response: The system displays the User Settings page.

3.1.5.3. Associated Functional Requirements

3.1.5.3.1. Volume Adjustment (O: Thomas)

The system shall implement a slider allowing the learner to adjust the volume of audio.

3.1.5.3.2. Volume Mute (O: Thomas)

The system shall implement a toggle allowing the learner to enable and disable audio.

3.1.5.3.3. Display Scale (O: Keckley, M1: Thomas)

The system shall expose a drop-down element for the learner to select an integer scale at which the complete interface will be displayed.

3.1.6. Admin Panel

3.1.6.1. Introduction/Purpose of Feature

The Admin Panel allows an admin to configure KIP platform settings during service.

3.1.6.2. Stimulus/Response

Stimulus: The admin clicks the Back to Dashboard button.

Response: The system displays the Dashboard page.

3.1.6.3. Associated Functional Requirements

3.1.6.3.1. Settings Return (O: Elmousalami)

The system shall display the Dashboard page when the admin clicks the Back to Dashboard button on the Admin Panel.

3.1.7. Sitewide Theme

3.1.7.1. Introduction/Purpose of Feature

KIP presents a consistent theme across every page.

3.1.7.2. Stimulus/Response

Stimulus: The user navigates to a KIP page.

Response: The system displays the page using the sitewide theme.

3.1.7.3. Functional Requirements

3.1.7.3.1. Button Consistency (O: Elmousalami, M1: Keckley)

The system shall display all buttons using a consistent visual style.

3.1.7.3.2. Content Title (O: Elmousalami)

The system shall display the title of each dashboard box in a title bar above the content.

3.1.7.3.3. Integer Scaling (O: Keckley)

The system shall display all visual components at scales of integer multiples to avoid the visual distortion caused by the non-integer scaling of low-resolution assets.

3.1.7.3.4. Color Palette (O: Keckley)

The system shall display user interface elements using a consistent predefined color palette.

3.1.7.3.5. Tiled Background (O: Keckley, M1: Elmousalami)

The system shall display a tiled background asset underneath all user interface elements across every page.

3.1.7.3.6. Reusable GUI Button (O: Noor)

The system shall provide a reusable GUI button with configurable caption, color, and dimensions using a 9-slice tiling scheme.

3.1.7.3.7. Reusable GUI Window (O: Noor)

The system shall provide a reusable GUI window with configurable title and dimensions.

3.1.8. Game Window

3.1.8.1. Introduction/Purpose of Feature

The Game Window displays the current scenario within a learner's active game session. The learner reviews the scenario and then chooses a response.

3.1.8.2. Stimulus/Response

Stimulus: The learner starts a game session.

Response: The system displays the game window with the first scenario.

Stimulus: The learner chooses an evidence item.

Response: The system displays the details of that evidence item.

Stimulus: The learner activates the exit control.

Response: The system closes the game window.

3.1.8.3. Associated Functional Requirements

3.1.8.3.1. Game Title

The system shall display the name of the active game as the title of the game window.

3.1.8.3.2. Exit

The system shall provide the capability for the learner to exit the active game session.

3.1.8.3.3. Scenario Topic

The system shall display the topic of the current scenario.

3.1.8.3.4. Scenario Progress Markers

The system shall display one marker for each scenario in the game session.

3.1.8.3.5. Scenario Description

The system shall display the situation description of the current scenario.

3.1.8.3.6. Evidence Item List

The system shall display the list of evidence items for the current scenario.

3.1.8.3.7. Evidence Item Detail

The system shall display the details of an evidence item when selected.

3.1.8.3.8. Response Choices

The system shall display the response choices for the current scenario.

3.1.9. Game Engine

3.1.9.1. Introduction/Purpose of Feature

The Game Engine manages active game sessions and controls the progression of scenarios and stages according to learner responses.

3.1.9.2. Stimulus/Response

Stimulus: A learner confirms a selected game from the browser to be started.

Response: The system loads the selected game to the web browser and configures the game.

3.1.9.3. Functional Requirements

3.1.9.3.1. Game Session Initialization (O: Keckley, M1: Flynn)

The system shall initialize the first scenario and its initial stage when a game session begins.

3.1.9.3.2. Response Submission (O: Keckley)

The system shall allow the learner to submit one of the response choices associated

with the current stage of the active scenario.

3.1.9.3.3. Stage Progression (O: Keckley)

The system shall advance the active scenario to the stage associated with the learner's selected response when that response does not terminate the scenario.

3.1.9.3.4. Scenario Completion (O: Keckley)

The system shall display the result of the current scenario when the learner selects a response that terminates the scenario.

3.1.9.3.5. Game Session Completion (O: Keckley)

The system shall complete the active game session when no additional scenarios remain.

3.1.9.3.6. Scenario Continuation (O: Keckley)

The system shall allow the learner to continue to the next scenario after completing the current scenario when additional scenarios remain in the active game session.

3.1.9.3.7. Game Session Abandonment (O: Keckley)

The system shall abandon the active game session when the learner exits before completing the session.

3.1.10. Progress Tracking

3.1.10.1. Introduction/Purpose of Feature:

User progress is saved when a game is successfully completed.

3.1.10.2. Stimulus/Response

Stimulus: The Feedback system sends completed progression data to the Progress Tracker.

Response: The system updates the learner's concept, topic, and game progress using the received data.

3.1.10.3. Functional Requirements

3.1.10.3.1. Concept Progress Update (O: Flynn)

The system shall acquire all data related to the current concept being saved for the user and update it with new information.

3.1.10.3.2. Topic Progress Update (O: Flynn)

The system shall acquire all data related to the current topic being saved for the user and update it with new information.

3.1.10.3.3. Game Progress Update (O: Flynn)

The system shall acquire all data related to the current game being saved for the user and update it with new information.

3.1.10.3.4. Completed Games History (O: Noor)

The system shall display a list of games completed by the user so the user can review their progress.

3.1.11. Curation

3.1.11.1. Introduction/Purpose of Feature

A learner is recommended topics according to the learner's demonstrated experience with them.

3.1.11.2. Stimulus/Response

Stimulus: A learner logs into the dashboard.

Response: The system displays recommendations in the dashboard under the recommendations section.

3.1.11.3. Functional Requirements

3.1.11.3.1. Trend: Making Progress (O: Keckley, Flynn)

The system shall recommend one topic that the learner has shown improvement in.

3.1.11.3.2. Trend: Needs Practice (O: Keckley, Flynn)

The system shall recommend one topic that the learner has shown little competency in.

3.1.11.3.3. Graphical Display (O: Flynn)

The system shall have visuals displayed on the web browser for the top recommendations.

3.1.12. Feedback

3.1.12.1. Introduction/Purpose of Feature

The Feedback system evaluates a learner's responses and provides immediate feedback during an active game session. Relevant response data shall be forwarded to Progress Tracking.

3.1.12.2. Stimulus/Response

Stimulus: The Game Engine submits a learner's response to the Feedback system.

Response: The system evaluates and records the response, returns the associated feedback, and forwards relevant progression data to the Progress Tracker.

3.1.12.3. Functional Requirements

3.1.12.3.1. Save Answer (O: Flynn)

The system shall acquire the question response data and mark it to its internal storage.

3.1.12.3.2. Send Data To Progress Updater (O: Flynn)

The system shall relay all answer data to the progress updater.

3.1.12.3.3. Feedback Response (O: Flynn)

The system shall return a message for a question’s or scenario’s response.

3.1.12.3.4. Scenario Result Feedback (O: Keckley, M1: Flynn)

The system shall return the feedback and outcome associated with the user's terminal response when a scenario is completed.

3.1.13. Legal Reference Algorithm

3.1.13.1. Introduction/Purpose of Feature

Legal content shall be automatically checked for any changes made to maintain accuracy.

3.1.13.2. Stimulus/Response

Stimulus: The system automatically invokes the Legal Reference Algorithm when a legal reference is checked.

Response: The system checks the referenced legal source for changes.

3.1.13.3. Functional Requirements

3.1.13.3.1. Automatic Invocation (O: Flynn)

The system shall automatically invoke the Legal Reference Algorithm when a legal reference is checked.

3.1.13.3.2. Change Detection (O: Flynn)

The system shall check for changes in the legal source material.

3.1.13.3.3. Change Indication (O: Flynn)

The system shall show if any changes were made to the legal source material in the database.

3.1.14. Data Initialization Layer and Metric Tracking

3.1.14.1. Introduction/Purpose of Feature

The Data Initialization Layer automatically seeds core database tables during application startup to support runtime simulation, testing, and prototype development. This layer uses JHipster's built-in Liquibase engine to parse static custom data. The application reads pre-configured .csv files via custom .xml changelog configuration files, delivering datasets to the frontend Dashboard feature.

3.1.14.2. Stimulus/Response

Stimulus: The system starts the KIP application via ./gradlew wrapper.

Response: System parses directory /fake-data, validating foreign keys and populating the tables with the custom .csv files delimited by semicolon.

3.1.14.3. Functional Requirements

3.1.14.3.1. Delimiter Enforcement (O: Thomas)

The system shall parse fake-data assets using the semicolon delimiter specified by their XML configuration files to prevent commas within legal text from being interpreted as column delimiters.

3.1.14.3.2. Topic Progress Integrity (O: Flynn, M1: Thomas)

The system shall map topic tracking metrics exclusively to track parent registries in order to display Recently Played.

3.1.14.3.3. Scenario Progress Constraints (O: Flynn, M1: Thomas)

To adhere to database entity requirements, the Scenario Progress .csv shall enforce a minimum validation rule of five questions for parameter, maxQuestions. This allows each tracking profile sufficient iterations for student evaluation data.

3.1.15. User Profile and Match Metrics Initialization

3.1.15.1. Introduction/Purpose of Feature

The user profile and match metrics layer maintains learner profile information and game activity.

3.1.15.2. Stimulus/Response

Stimulus: An Authenticated user logs into the application.

Response: The system loads corresponding user profile data alongside cumulative match metrics to display performance summaries.

3.1.15.3. Functional Requirements

3.1.15.3.1. Identity Binding (O: Thomas)

The system shall associate each custom application UserProfile entity directly with a built-in security user account ID.

3.1.15.3.2. Match Sessions Played (O: Flynn, M1: Thomas)

The system shall track the total number of completed game sessions per profile. This data type shall be non-negative integers.

3.1.15.3.3. Performance Range Validation (O: Flynn, M1: Thomas)

The system shall store the user’s score average as a floating-point value between 0.00 (0%) and

1.00 (100%), inclusive.